DATA SCIENCE & ANALYTICS

MODULE 03 SESSION 13

Groups without a label: k-means clustering

What kinds of shoppers do we have, when no column records it?

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BY THE END OF TODAY, YOU CAN

Class objectives

01 Train KMEANS and read its groups

02 Explain distance and one round of k-means

03 Explain why scaling and logs fix different problems

04 Profile groups in plain units; read Davies–Bouldin

05 Choose k, check the groups, give each an action

EVERYONE DO

Step 1.1

Warm-up

Five questions on D12. A1 was due today.

Suggested time · 10 minutes

PART 01

A first clustering, before the theory

200 mall shoppers: upload, look, run a prepared k-means model, look at its groups.

INSTRUCTOR DO

Step 2.1

Today’s question

A mall, its card members, and no answer column.

Suggested time · 4 minutes

What kinds of shoppers do we have, and does each kind deserve a different campaign?

FIVE OF 200 CARD MEMBERS · MALL CUSTOMERS

No column says which kind of shopper

customer_id	gender	age	annual_income_k	spending_score
1	Male	19	15	39
2	Male	21	15	81
3	Female	20	16	6
4	Female	23	16	77
5	Female	31	17	40

Income: thousands of dollars a year. Spending score: 1 to 100, set by the mall from how often and how much the member buys.

EVERYONE DO

Step 2.2

Upload and look

Count the shoppers, check the two columns we will compare, then look.

Suggested time · 12 minutes

labs/D13/01-Evr-First-Segments/README.md

Two columns on similar scales

MIN and **MAX**: the range of each column we will compare.

```
SELECT COUNT(*) AS shoppers,
       MIN(annual_income_k), MAX(annual_income_k),
       MIN(spending_score), MAX(spending_score)
FROM ml.mall;
```

→ 200 · income 15 to 137 · score 1 to 99

200 SHOPPERS · INCOME AGAINST SPENDING SCORE

How many groups do you see?



One dot is one shopper. The model will see only these two numbers per shopper.

EVERYONE DO

Step 2.3

A prepared k-means model, and its groups

Train, count each group, read three shoppers from each.

Suggested time · 15 minutes

labs/D13/01-Evr-First-Segments/README.md

BLOCK 3 · TRAIN

Five groups, from two columns

`model_type = 'KMEANS'`: clustering. No label. We choose k = 5.

```
CREATE OR REPLACE MODEL ml.mall_segments
OPTIONS (
  model_type = 'KMEANS',
  num_clusters = 5,
  kmeans_init_method = 'KMEANS++',
  standardize_features = TRUE
) AS
SELECT annual_income_k, spending_score
FROM ml.mall;
```

DEAD END · SELECT * IN CREATE MODEL

Every selected column is used to measure distance

Column	With SELECT *	Why that is wrong
customer_id	**a feature**	**shopper 1 and shopper 200 look far apart because of card numbers**
age, gender	features	not what marketing asked us to compare
annual_income_k, spending_score	features	the two we chose

So the SELECT lists exactly the columns that define “alike”. Choosing them is part of the model.

Count the shoppers in each group

centroid_id: the group joined. A label, not a rank.

```
SELECT centroid_id, COUNT(*) AS shoppers
FROM ML.PREDICT(MODEL ml.mall_segments,
       (SELECT * FROM ml.mall))
GROUP BY centroid_id
ORDER BY shoppers DESC;
```

→ most runs: 81, 39, 35, 23, 22

A group of about 100? Keep it; we compare runs later.

THE LOWEST-DISTANCE RUN · k = 5

Five groups, as the model drew them



Colour: the group. Diamond: the centre of each group. Your group numbers will differ; the shapes should not.

INSTRUCTOR DO

Step 2.4

What did it just do?

How did it decide who goes together? Why five? Are these the right groups?

Suggested time · 4 minutes

PART 02

How k-means makes groups

No label, distance, one round at a time, a round by hand, and what BigQuery stores.

INSTRUCTOR DO

Step 3.1

Learning without a label

D11’s map of the kinds of learning, with today lit.

Suggested time · 7 minutes

KINDS OF MACHINE LEARNING

Four kinds, told apart by what the rows hold

Supervised: a salary (D11), churn Yes or No (D12).

Kind	The rows hold	Typical tasks
Supervised	**features and the label**	**regression, classification**
Unsupervised	features only	clustering, fewer columns (PCA), anomalies
Semi-supervised	a few labeled rows, many unlabeled	label a little, learn from all
Reinforcement	actions and rewards	an agent learning by trial

KINDS OF MACHINE LEARNING

Four kinds, told apart by what the rows hold

Unsupervised: today’s shopper groups; unusual clients in lab 08.

Kind	The rows hold	Typical tasks
Supervised	features and the label	regression, classification
Unsupervised	**features only**	**clustering, fewer columns (PCA), anomalies**
Semi-supervised	a few labeled rows, many unlabeled	label a little, learn from all
Reinforcement	actions and rewards	an agent learning by trial

DEFINITION

Clustering puts rows into groups so that rows in a group are more alike than rows in different groups. There is no target column.

Nobody tells the model what the groups are; a person names them afterward.

The column that disappears

Day	One row	Label	Kind
D11	a past hire	salary: a number	supervised: regression
D12	a telecom customer	Churn: Yes or No	supervised: classification
D13	a card member	**none**	**unsupervised: clustering**

INSTRUCTOR DO

Step 3.2

Distance between two shoppers

Close in the measurements means a short straight line.

Suggested time · 8 minutes

EUCLIDEAN DISTANCE · BIGQUERY'S DEFAULT

Shoppers 3 and 124 are 100.2 apart



√(53² + 85²) = √(2,809 + 7,225) = 100.2. Shopper 11 (19, 14) is only 8.5 from shopper 3.

INSTRUCTOR DO

Step 3.3

k-means, one round at a time

Assign, update, repeat, on the 200 shoppers.

Suggested time · 10 minutes

DEFINITION

k-means chooses k centers, then repeats two steps until nothing changes: assign each row to its nearest center; move each center to the mean of its rows.

The centers are called centroids. Rows must be numeric and on comparable scales.

Start: five centres at random shoppers



Mean squared distance to the nearest centre: 1.07 (scaled units).

Assign: each shopper joins its nearest centre



Colour shows the centre each shopper joined. Still 1.07.

Update: each centre moves to its group’s average



Arrows: how far each centre moved. Now 0.63.

After 5 rounds, nothing moves



Mean squared distance 0.33. Training stops when a round barely helps.

STUDENT DO

One round by hand

Six shoppers, two centres, a deliberately poor start. Assign, then update.

Suggested time

12 minutes

`labs/D13/02-Stu-One-Round-By-Hand/README.md`

REVIEW

Time's up.
Let's review.

One round by hand

Solved: labs/D13/02-Stu-One-Round-By-Hand/Solved/one_round.md

START A (15, 81) · B (17, 76)

One shopper against five

Shopper	To A	To B	Joins
2 (15, 81)	0.0	5.4	A
4 (16, 77)	4.1	1.4	B
6 (17, 76)	5.4	0.0	B
124 (69, 91)	54.9	**54.1**	B
126 (70, 77)	55.1	53.0	B
128 (71, 95)	57.7	57.2	B

Update: A stays at (15, 81); B moves to (48.6, 83.2), into the empty space between the two real groups.

EVERYONE DO

Step 3.6

What BigQuery stored: centres and distances

The centre of each group, and each shopper’s distance to every centre.

Suggested time · 12 minutes

labs/D13/01-Evr-First-Segments/README.md

The centres, in thousands of dollars and points

ML.CENTROIDS: 10 rows, returned unscaled by default.

```
SELECT centroid_id, feature,
       ROUND(numerical_value, 1) AS centre
FROM ML.CENTROIDS(MODEL ml.mall_segments)
ORDER BY centroid_id, feature;
```

→ (26, 79) (26, 21) (55, 50) (87, 82) (88, 17)

Each shopper joins the nearest centre

nearest_centroids_distance: a list per shopper, nearest first, in scaled units.

```
SELECT customer_id, centroid_id, nearest_centroids_distance
FROM ML.PREDICT(MODEL ml.mall_segments,
       (SELECT * FROM ml.mall WHERE customer_id IN (3, 124)));
```

→ 2 rows · shopper 3: nearest 0.70 · 124: 0.75

PART 03

Units, scaling and logs

Grams against millimetres, a heavy tail, and the two fixes.

INSTRUCTOR DO

Step 4.1

Grams against millimetres

Two penguins, four measurements, one distance.

Suggested time · 8 minutes

AN ADELIE AND A GENTOO · PALMER PENGUINS

The grams take over the distance

Measurement	Difference	Squared	Std dev	Scaled diff
bill length (mm)	6.1	37	5.5	1.12
bill depth (mm)	2.9	8	2.0	1.47
flipper (mm)	34	1,156	14.1	2.42
body mass (g)	1,550	**2,402,500**	802	1.93
distance		**1,550.4**		**3.60**

Standardising divides each difference by its column’s standard deviation: 800 g and 5 mm each count as about one step. BigQuery’s default.

EVERYONE DO

Step 4.2

The penguins, with and without scaling

Three groups twice, compared with the species the model never sees.

Suggested time · 20 minutes

labs/D13/03-Evr-Scale-The-Penguins/README.md

BLOCK 2 · SCALING SWITCHED OFF

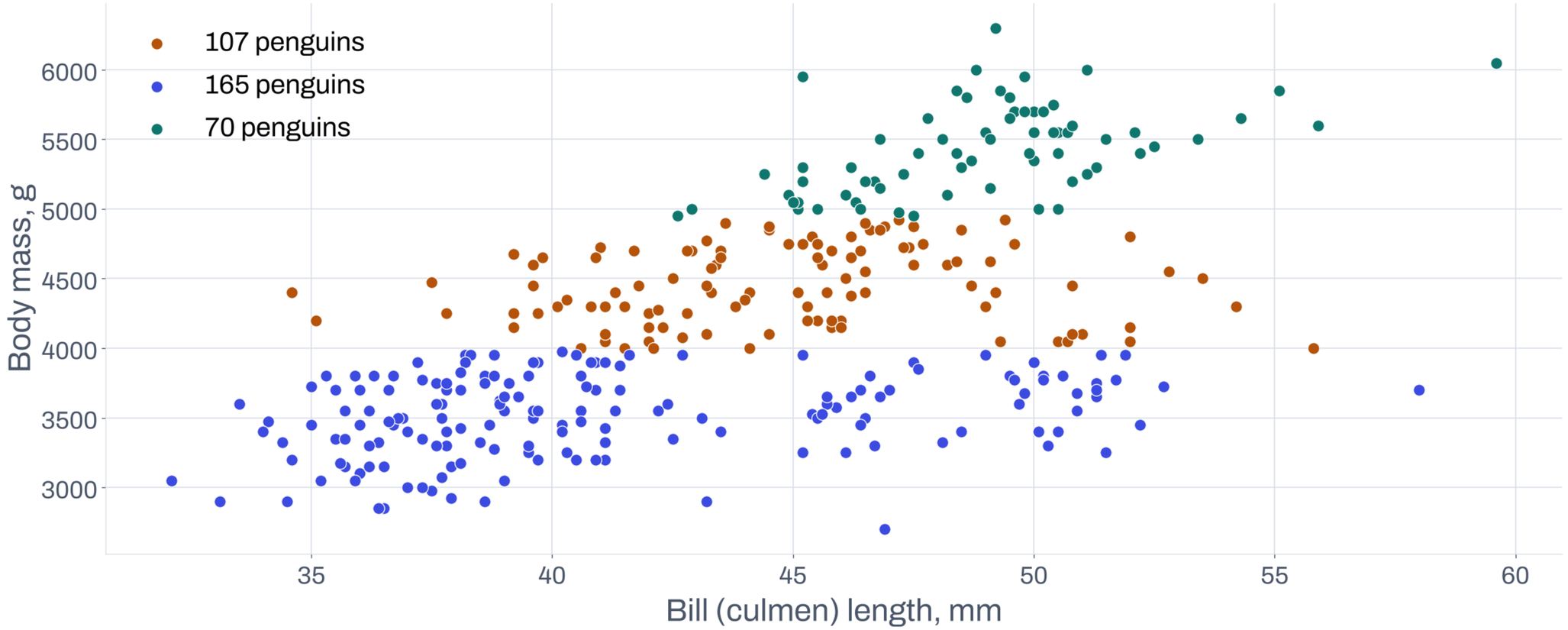
Four measurements as they are

`standardize_features = FALSE`: grams and millimetres go in unchanged.

```
CREATE OR REPLACE MODEL ml.penguins_raw
OPTIONS (
  model_type = 'KMEANS', num_clusters = 3,
  kmeans_init_method = 'KMEANS++',
  standardize_features = FALSE
) AS
SELECT culmen_length_mm, culmen_depth_mm,
       flipper_length_mm, body_mass_g
FROM `bigquery-public-data.ml_datasets.penguins`
WHERE body_mass_g IS NOT NULL;
```

k = 3 · NO SCALING · 342 PENGUINS

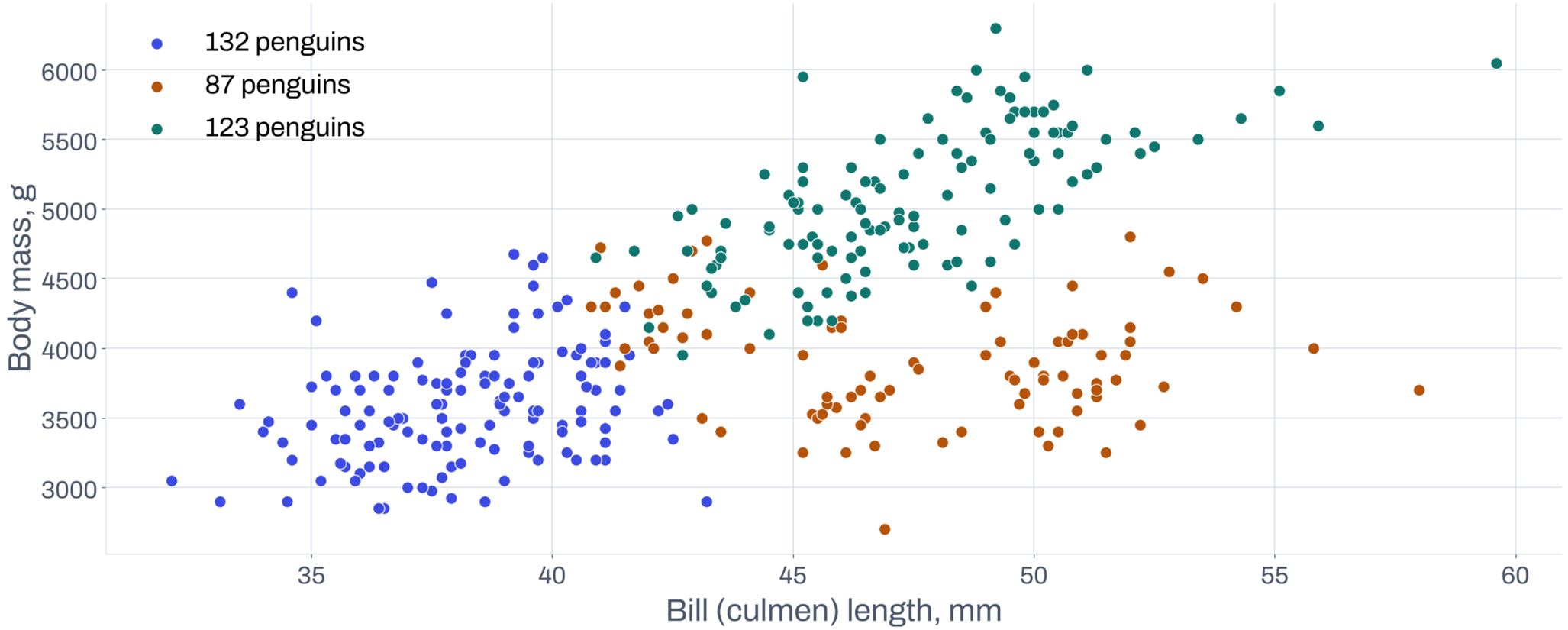
Without scaling: cut by body mass



Groups: 2,700–3,975 g, 4,000–4,925 g, 4,950–6,300 g. Sizes vary by run (165 · 107 · 70, 170 · 111 · 61).

k = 3 · SCALED · 342 PENGUINS

With scaling: close to the three species



The same four measurements, each in standard deviations. Colour: the group the model found.

EACH GROUP AGAINST SPECIES · ONE RUN OF EACH

Scaled groups match species far better

	Adelie	Chinstrap	Gentoo
No scaling · lightest	112	52	1
No scaling · middle	39	16	52
No scaling · heaviest	0	0	70
Scaled · group 1	127	5	0
Scaled · group 2	24	63	0
Scaled · group 3	0	0	**123**

In a group where their species is the most common: about 7 in 10 penguins unscaled, 9 in 10 scaled.

THE ANSWER

Not wrong. k-means groups penguins that measure alike; species is a fact about the birds. A clustering summarises the columns you give it.

INSTRUCTOR DO

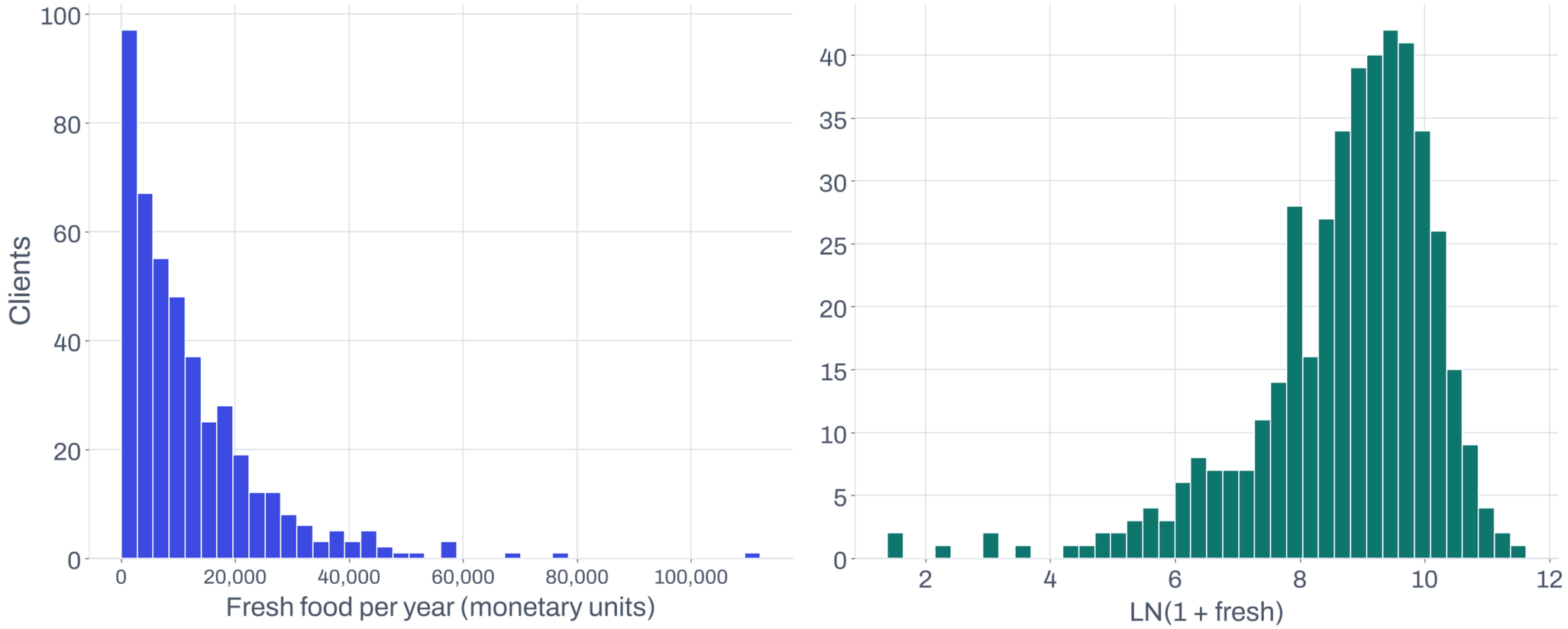
Step 4.3

Logs fix a different problem

A wholesaler’s 440 clients: most spend little, a few spend a lot.

Suggested time · 9 minutes

A heavy tail, and the same clients after LN(1 + x)



Left: as recorded. Right: LN(1 + fresh). Ten times more spending adds 2.3 at any size.

TWO PROBLEMS, TWO FIXES

Scaling fixes units; logs fix a long tail

Problem	Fix	Wholesaler, k = 4
columns in different units	scaling (the default)	one group of about 300; tiny groups of giants
a few huge values in a column	**LN(1 + x), then scaling**	four usable groups, 20 clients or more

INSTRUCTOR DO

Step 4.4

Ideas that come back on every ML problem

Scaling, k as a setting we choose, and randomness.

Suggested time · 5 minutes

NOT JUST K-MEANS

Three ideas you will meet again

Idea	Today	Also
scaling	grams against millimetres	regression weights (D11 lab 06), D14
a hyperparameter	k, chosen by us	L2 (D11), tree depth (D14)
randomness	random starting centres	random splits (D11, D12): run more than once

REVIEW

Time's up. Let's review.

Scaling and logs

Solved: labs/D13/03-Evr-Scale-The-Penguins/Solved/scale_the_penguins.sql

Break

30 minutes

After the break: with no test score, how do we judge five groups?

PART 04

Judging a clustering with no test score

Profiles, a score for the shape, k, a second run, and one action per group.

INSTRUCTOR DO

Step 5.1

No test score: what can we check?

Five checks instead of one score.

Suggested time · 5 minutes

Five checks, in this order

#	Check	How
1	Is each group different, in plain units?	a profile: join ML.PREDICT to the table
2	Are the groups tight and apart?	Davies–Bouldin index (ML.EVALUATE)
3	Would another k serve better?	compare k, then ask the business
4	Do the groups come back?	train again; compare
5	Does each group get its own action?	name it, give it one action

EVERYONE DO

Step 5.2

Profiles in original units

Size, the two columns, and the columns the model never used.

Suggested time · 12 minutes

labs/D13/04-Evr-Judge-The-Segments/README.md

Average each group in plain units

GROUP BY centroid_id: one row per group, any column you like.

```
SELECT centroid_id, COUNT(*) AS shoppers,
  ROUND(AVG(annual_income_k), 1) AS avg_income_k,
  ROUND(AVG(spending_score), 1) AS avg_score,
  ROUND(AVG(age), 1) AS avg_age,
  ROUND(COUNTIF(gender = 'Female') / COUNT(*), 2) AS share_women
FROM ML.PREDICT(MODEL ml.mall_segments, (SELECT * FROM ml.mall))
GROUP BY centroid_id ORDER BY avg_income_k;
```

FIVE GROUPS · SORTED BY INCOME

Five profiles marketing can read

Shoppers	Income (k$)	Score	Age	Share women
≈ 22	26	79	**25**	0.59
≈ 23	26	21	45	0.61
≈ 80	55	50	43	0.59
≈ 39	87	82	33	0.54
≈ 35	88	17	41	0.46

ML.CENTROIDS gave only income and score. The profile adds size and the columns the model did not use.

EVERYONE DO

Step 5.3

Davies–Bouldin, and k as a business choice

A score for the shape of the groups, across k.

Suggested time · 10 minutes

labs/D13/04-Evr-Judge-The-Segments/README.md

Two numbers, neither a test score

davies_bouldin_index: lower means tighter, better-separated groups.

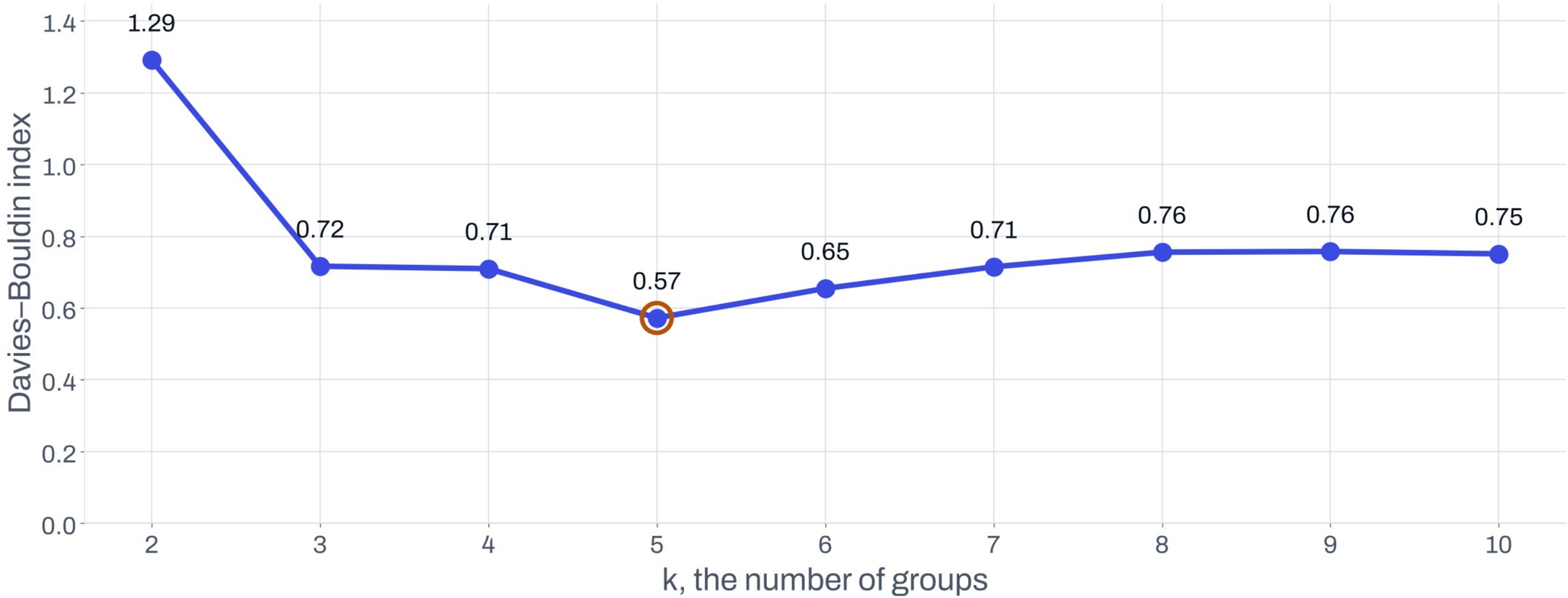
```
SELECT ROUND(davies_bouldin_index, 3) AS davies_bouldin,
       ROUND(mean_squared_distance, 3) AS mean_squared_distance
FROM ML.EVALUATE(MODEL ml.mall_segments);
```

→ Davies-Bouldin ≈ 0.57 · mean squared distance ≈ 0.33

Mean squared distance always falls as k grows: 0 with 200 groups of one.

THE LOWEST-DISTANCE RUN FOR EACH k

Lowest at k = 5; the business still chooses



k = 6 scores 0.65 and gives a ten-person group. A mall that wants a VIP list might still choose it.

INSTRUCTOR DO

Step 5.4

Run it again: stability

Back to the sizes you posted at the start.

Suggested time · 6 minutes

THE SAME STATEMENT · DIFFERENT RANDOM STARTS

Two kinds of answer

Sizes	Mean sq. distance	Runs, of 1,000	What it is
81 · 39 · 35 · 23 · 22 (or 80 · 39 · 36)	**0.33**	about 650	the five groups
100 · 39 · 28 · 23 · 10 and similar	about 0.49	about 350	stuck: a worse split

Run more than once. Keep the lowest mean squared distance, then check its profile still makes sense.

PARTNER DO

Name the segments, one action each

Marketing can run one campaign per group. A name a manager understands, and one action, for each.

Suggested time

12 minutes

labs/D13/04-Evr-Judge-The-Segments/README.md

PARTNER DO

Five names, five actions

INSTRUCTIONS	Use the five profiles: size, income, score, age. For each group write a short name that says what its members do, and one action marketing could take. Post five lines in Slack.
HINT	Name by behaviour: “high income, spends little here”, not “group 4”.
BONUS	Which group would you spend the first campaign on, and why?

Full steps and Check yourself: labs/D13/04-Evr-Judge-The-Segments/README.md

REVIEW

Time's up. Let's review.

Name the segments

Solved: labs/D13/04-Evr-Judge-The-Segments/Solved/judge_the_segments.sql

A SAMPLE ANSWER

Five groups, five actions

Group	Name	One action
≈ 22 · 26k · 79 · age 25	young big spenders	social-media offers, new arrivals
≈ 23 · 26k · 21 · age 45	careful, low budget	essentials and discount days
≈ 80 · 55k · 50 · age 43	the middle	the standard loyalty programme
≈ 39 · 87k · 82 · age 33	well-off big spenders	premium tier; keep them
≈ 35 · 88k · 17 · age 41	well-off, spend elsewhere	ask why; a first-visit offer

THE RULE

k-means groups customers by distance, so scale first. The charts suggest k, the business chooses it, and a person names the groups.

STUDY LIST · IN CLASS AND IN THE LABS

BigQuery ML functions today

In class	What it gives you	Activity
ML.PREDICT	each row’s group (centroid_id) and its distances	01, 03, 04
ML.CENTROIDS	the centre of each group, feature by feature	01
ML.EVALUATE	Davies–Bouldin index and mean squared distance	04

New in the labs	What it gives you	Lab
LN(1 + x)	a SQL function that tames a heavy tail before clustering	07
ML.TRAINING_INFO	each round: how far rows sit from their centres	06
ML.DETECT_ANOMALIES	flags rows far from their centre; billed even in a sandbox	08

EVERY MODULE 3 FUNCTION · resources/bigquery-ml-functions.md

CLOSE

What you can do now

- Train KMEANS; read its groups
- Explain distance and a round of k-means
- Tell scaling from logs
- Judge groups: profiles, Davies–Bouldin, a rerun

NEXT · D14 Improving models.

LAB

AFTER CLASS · SELF-GUIDED

Clustering, one idea per lab

TASK

Four short labs: choosing k; stable groups; logs; odd ones out.

FILES

`labs/D13/05-Lab-Choosing-K-And-Tuning/`

`labs/D13/06-Lab-Starts-And-Stopping/`

`labs/D13/07-… and 08-Lab-Odd-Ones-Out/`

DONE WHEN

Each lab ends with a table of what you saw.

Optional self-study: lab 09, Nakheel’s customers by k-means.